KUNAL CHAND

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SUMMARY

AI Full Stack Software Engineer with 5 years of experience building scalable, production-grade applications across AI-driven and distributed systems. Specialized in developing LLM-powered solutions, multi-agent architectures, and retrieval-based pipelines for intelligent automation. Strong background in full-stack development using modern frontend and backend frameworks, along with cloud-native deployment and microservices design. Experienced in healthcare data systems, real-time processing, and secure, compliant architectures, with a focus on delivering reliable, high-performance software solutions.

CORE SKILLS

Programming Languages: Python, Java, JavaScript (ES6+), TypeScript, SQL, C++, Go, Bash

AI / Machine Learning: Large Language Models (LLMs), GPT-4/4o, Claude API, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), Multi-Agent Systems, Prompt Engineering, Fine-tuning, Reinforcement Learning from Human Feedback (RLHF), BERT, Transformer Architectures, Model Evaluation & Optimization

Generative AI & NLP: LangChain, LangGraph, LlamaIndex, Hugging Face Transformers, Sentence-Transformers, Tokenization, Embeddings, Semantic Search, Named Entity Recognition (NER), Text Classification, Document Intelligence Pipelines

Frontend Development: React.js, Next.js, Redux Toolkit, React Router, Zustand, Material UI, Tailwind CSS, Chart.js, D3.js, HTML5, CSS3, Responsive UI Design, WebSockets

Backend Development: FastAPI, Node.js, Express.js, Spring Boot, RESTful APIs, GraphQL, Microservices Architecture, API Gateway Design, Server-Sent Events (SSE), Celery, Redis

Data Engineering: Apache Airflow, Kafka, ETL/ELT Pipelines, Data Ingestion, Data Transformation, Batch & Stream Processing

Databases & Storage: PostgreSQL, MongoDB, MySQL, OracleDB, Microsoft SQL Server, Neo4j (Graph DB), Redis, ChromaDB, Elasticsearch, Data Modeling, Query Optimization

Cloud & DevOps: AWS (EC2, S3, Lambda, RDS, ECR, SageMaker, SQS), Azure OpenAI, Google Cloud Platform (GCP), Docker, Kubernetes, Terraform (IaC), CI/CD (GitHub Actions, Jenkins), API Gateway, Autoscaling, Blue-Green Deployments

MLOps & Observability: Weights & Biases, Prometheus, Grafana, Logging, Model Monitoring & Experiment Tracking

System Design & Architecture: Distributed Systems, Scalable System Design, Event-Driven Architecture, High Availability Systems, Performance Optimization

Testing & Quality Engineering: Test-Driven Development (TDD), Unit Testing (JUnit, Mockito), Integration Testing, Test Automation, SonarQube, Code Quality & Coverage

Collaboration & Tools: Git, GitHub, Agile/Scrum, Jira, Postman

PROFESSIONAL EXPERIENCE

AI Software Engineer | Gradient Certification Inc, New York, USA — **Feb 2026 – Present**

- Designed and implemented a multi-agent AI orchestration system using LangGraph and GPT-4o to automate compliance workflows, including document intake, classification, and regulatory requirement mapping across ISO-based certification processes.
- Built a RAG-driven pipeline using ChromaDB and LangChain to extract, summarize, and validate regulatory content from 10,000+ client documents, improving audit readiness and reducing manual review effort.
- Developed a full-stack compliance platform using Next.js and FastAPI, enabling real-time AI-assisted audit report generation, progress tracking, and gap analysis with streaming LLM responses (SSE).
- Engineered AI-powered validation and integrity checks using sentence-transformers to detect missing, inconsistent, or non-compliant information, enhancing accuracy in certification assessments and audit reporting.

Full Stack Software Engineer | Trulogik Inc, Maryland, USA — **Jul 2024 – Nov 2025**

- Led the end-to-end development of a HIPAA-compliant claims adjudication platform using React.js, FastAPI, and PostgreSQL, automating the intake and routing of 50,000+ monthly insurance claims and reducing manual review effort by 62%.
- Built a multi-stage LLM pipeline with LangChain and GPT-4 to extract structured clinical data (ICD-10, CPT codes, and billing modifiers) from unstructured EOBs and clinical notes, achieving 93% field-level accuracy.
- Developed a semantic similarity engine using FAISS and OpenAI embeddings to identify comparable historical claims, improving decision consistency and reducing adjudication variance by 41%.
- Designed a real-time HL7 FHIR R4 ingestion microservice using Node.js and Kafka to process payer EHR claim feeds, normalize data into a unified schema, and store it in MongoDB with full audit tracking, achieving sub-3-second latency.

- Implemented a BERT-based classification model trained on 200K+ annotated claims to detect denial reasons and coverage conflicts, delivering a 91% macro F1 score and improving claim categorization accuracy.
- Engineered a Python-based adjudication rules engine with 300+ CMS and payer rules, paired with a React interface for non-technical users, and leveraged Redis, Celery, and AWS SQS to handle 5x traffic spikes while maintaining 99.9% SLA.

Research Software Engineer | University at Buffalo, New York, USA **Jan 2024 – Jul 2024**

- Built a large-scale knowledge graph platform using Neo4j, FastAPI, and React to map relationships across 500K+ academic papers, enabling semantic, cross-domain research discovery for faculty across 14 departments.
- Developed an LLM-driven document processing pipeline using LangChain, Hugging Face Transformers, and Weaviate to extract entities, methodologies, and findings from research papers, reducing manual literature review effort by 75%.
- Designed a multi-modal retrieval system integrating GPT-4 Vision and CLIP embeddings to support both text and image-based queries, allowing researchers to discover papers through diagram and concept similarity.
- Created a collaborative annotation and experiment tracking interface using Next.js, Zustand, PostgreSQL, and MLflow, enabling teams to link datasets with hypotheses and maintain full experiment lineage.
- Implemented a distributed ingestion pipeline with Apache Airflow and AWS S3 to automatically collect and process data from arXiv, PubMed, and Semantic Scholar APIs at a scale of 10K papers per day.
- Deployed and maintained the platform on AWS (EC2, RDS, SageMaker) using Terraform and API Gateway, achieving 99.95% uptime and supporting autoscaling to handle 3x traffic spikes during peak research periods.

Software Developer | Cognizant Technology Solutions (Client: Anthem Inc., Healthcare**), India** **Jan 2020 – Aug 2022**

- Developed a FHIR R4-compliant REST API gateway using Spring Boot and Java 11 to support prior authorization workflows across 12 hospital networks, reducing manual review by 38% while handling 50,000+ daily requests with sub-300ms response time.
- Built a real-time claims adjudication microservice integrated with ICD-10 and CPT code systems using Apache Kafka, processing over 3M monthly claims with sub-200ms latency and reducing adjudication errors by 31%.
- Designed and delivered a React.js-based provider portal with role-based access control using Spring Security and JWT, enabling 8,000+ clinicians to manage authorizations, track requests, and submit appeals through a unified interface.
- Created an automated eligibility verification engine by converting CMS Medicare/Medicaid SOAP feeds into RESTful services, reducing first-pass eligibility failures by 27% and improving plan resolution accuracy to 98.4%.
- Implemented a secure PostgreSQL audit trail system with AES-256 row-level encryption to meet HIPAA §164.312 compliance standards and support regulatory reporting requirements.
- Designed and deployed a Kubernetes-based batch reconciliation pipeline using Spring Batch to process EOB records across four regional data centers, reducing discrepancies by 41% compared to legacy COBOL systems.
- Integrated Twilio-based SMS and email notifications into the patient portal to deliver real-time authorization updates, achieving 91% engagement and reducing provider call center volume by 22%.

PROJECTS

PDF GenAI Chatbot | Python, LangChain, Vector Embeddings, Natural Language Processing (NLP)

- Developed a Retrieval-Augmented Generation (RAG) chatbot to analyze 100+ page PDFs and deliver context-aware answers using LangChain. Integrated Pinecone with HuggingFace embeddings to improve semantic search and retrieval accuracy, enabling efficient document understanding and interactive querying.

Timesheet Auto Filler | Java, Selenium WebDriver, OOP

- Developed a Java-based automation tool using Selenium and object-oriented design principles to streamline timesheet entry in an internal application. Implemented a one-click workflow that reduced manual input time from 10–15 minutes to under 2 minutes, improving daily productivity and minimizing repetitive effort.

EDUCATION

Master of Science in Computer Science and Engineering
University at Buffalo, The State University of New York, Buffalo, New York, USA

Bachelor of Engineering in Computer Science and Engineering
Kalinga Institute of Industrial Technology (KIIT University), Bhubaneswar, Odisha, India